

EXPLORATORY DATA ANALYSIS (EDA) ON TITANIC DATASET

ABSTRACT

This project focuses on performing Exploratory Data Analysis (EDA) on the Titanic dataset to identify patterns, trends, and relationships affecting passenger survival. The analysis uses statistical summaries, data cleaning techniques, and visualizations to better understand the dataset. Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used for analysis and graphical representation.

INTRODUCTION

Exploratory Data Analysis (EDA) is an important step in data science and machine learning. It helps in understanding the structure of data, identifying missing values, detecting patterns, and discovering useful insights before model building. The Titanic dataset is a well-known dataset containing passenger information such as age, gender, ticket fare, passenger class, and survival status.

OBJECTIVES

- To analyze the Titanic dataset.
- To identify missing values and clean the data.
- To visualize important trends and patterns.
- To study factors affecting passenger survival.
- To develop analytical thinking and data visualization skills.

TOOLS AND TECHNOLOGIES USED

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab / Jupyter Notebook

DATASET DESCRIPTION

The Titanic dataset contains information about passengers aboard the Titanic ship. It includes features such as Passenger ID, Age, Gender, Passenger Class, Fare, Embarked Port, and Survival Status.

Dataset Source:

<https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv>

METHODOLOGY

1. Import required Python libraries.
2. Load dataset from an online source.
3. Perform data cleaning and preprocessing.
4. Analyze statistical summaries.
5. Create visualizations using Matplotlib and Seaborn.
6. Generate insights based on analysis.

OUTPUT SCREENSHOTS

Dataset Preview and Shape

First 5 Rows of Dataset:

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

Shape of Dataset:
(891, 12)

Dataset Information and Missing Values

```

Column Names:
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
      'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')

```

```

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null   int64
1   Survived     891 non-null   int64
2   Pclass       891 non-null   int64
3   Name         891 non-null   object
4   Sex          891 non-null   object
5   Age          714 non-null   float64
6   SibSp        891 non-null   int64
7   Parch        891 non-null   int64
8   Ticket       891 non-null   object
9   Fare         891 non-null   float64
10  Cabin        204 non-null   object
11  Embarked     889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None

```

```

Missing Values:
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64

```

Statistical Summary and Data Cleaning

Missing Values:

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age           177
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin         687
Embarked       2
dtype: int64
```

Statistical Summary:

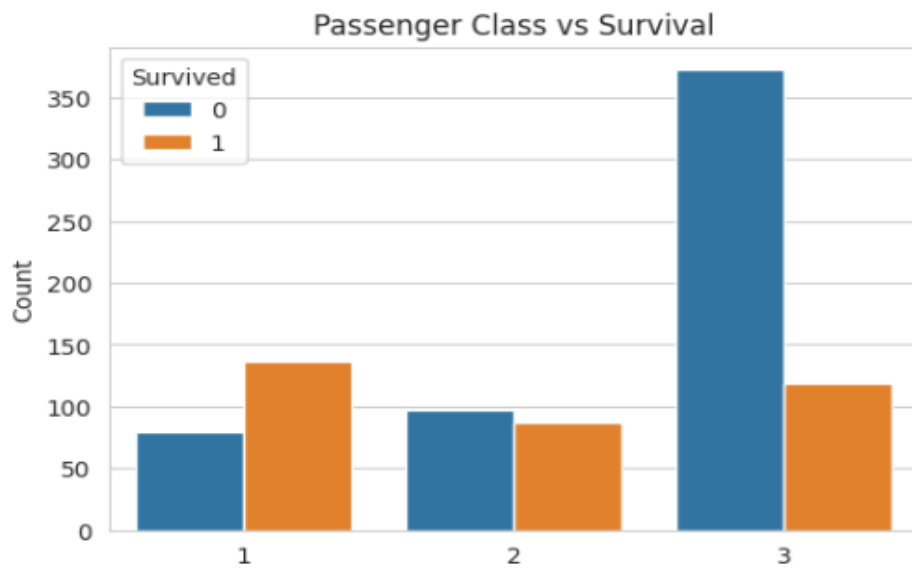
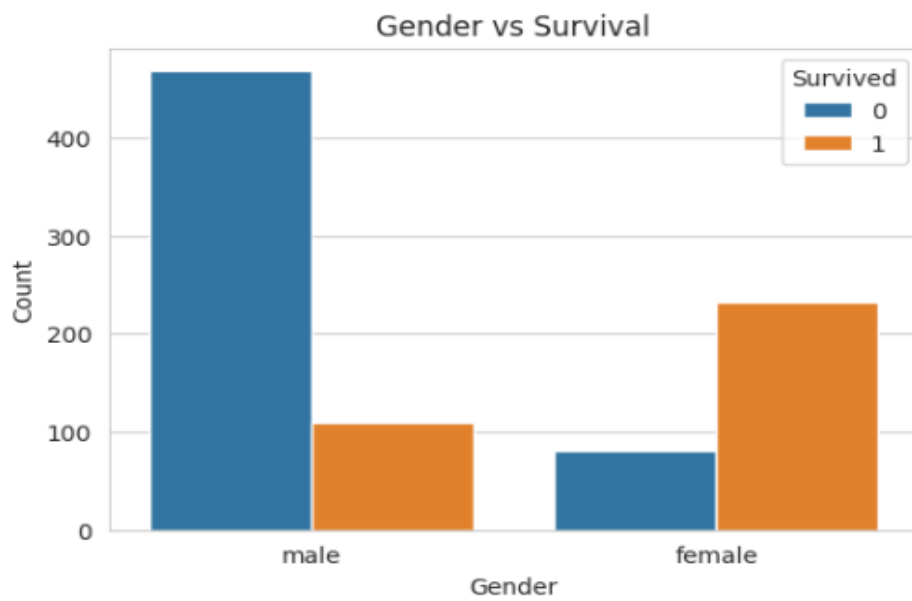
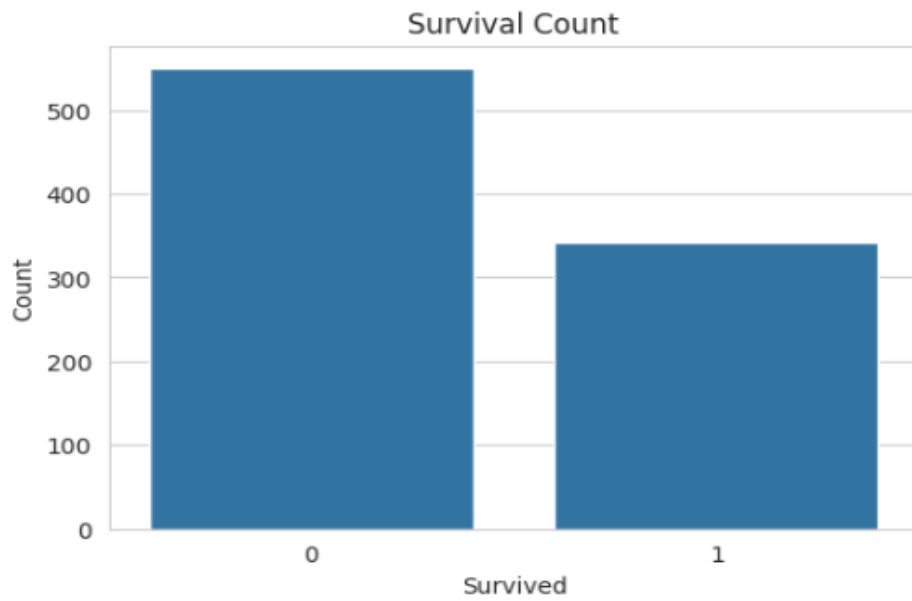
	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

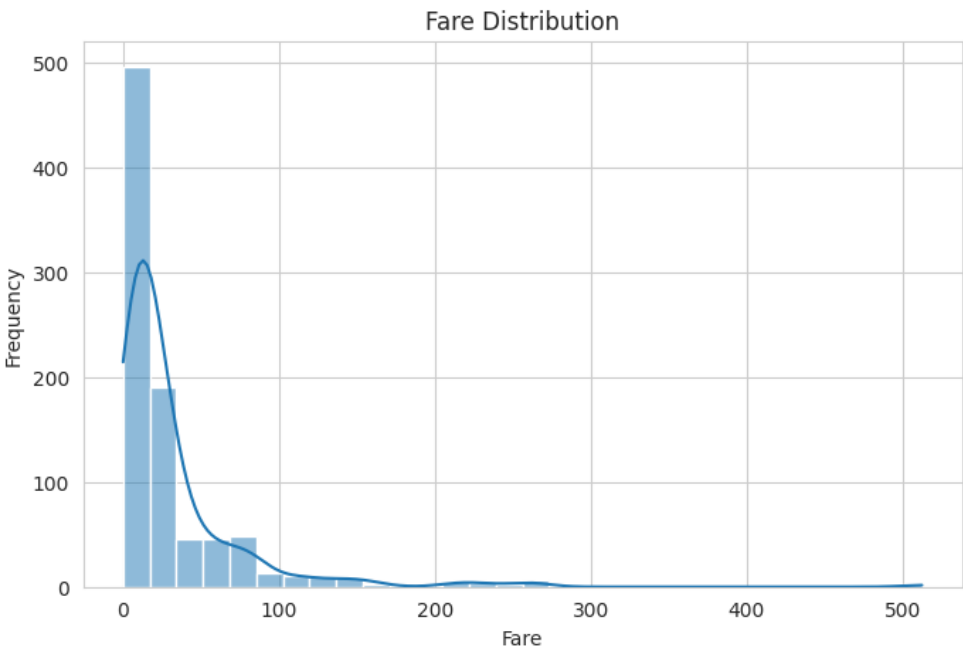
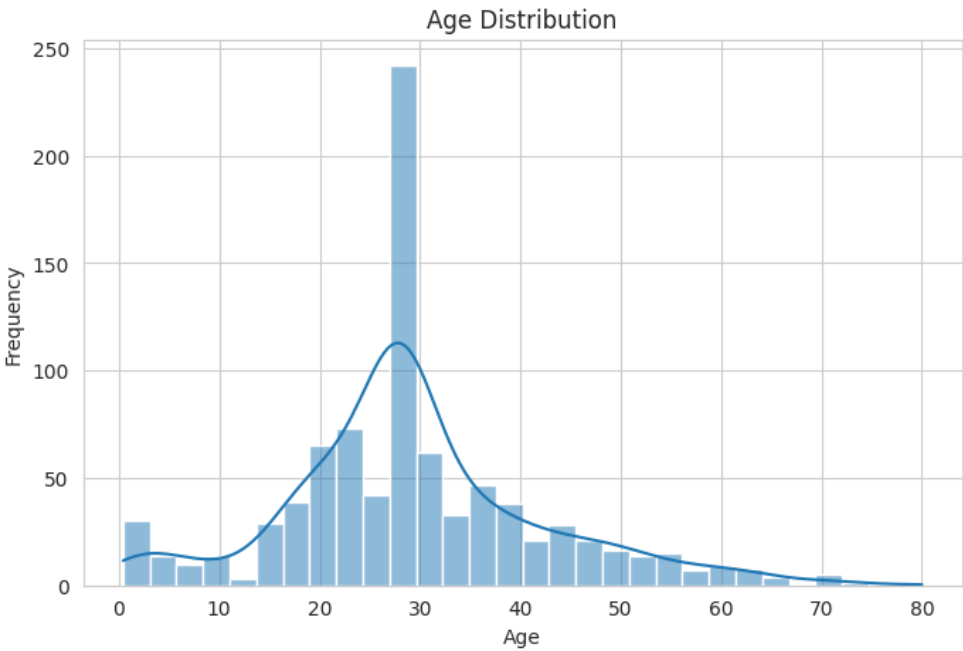
Missing Values After Cleaning:

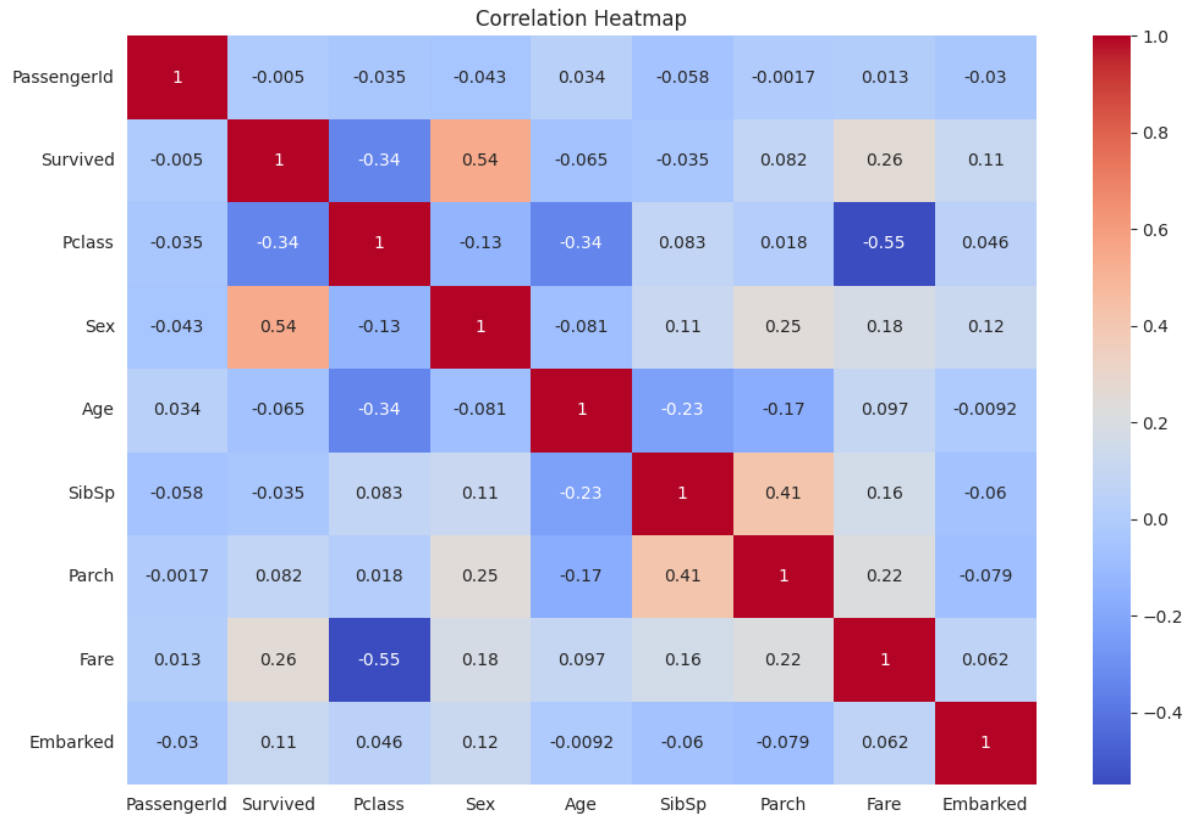
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

Survival and Passenger Analysis



Age and Fare Distribution





Survival Rate by Gender:

Sex

female 0.742038

male 0.188908

Name: Survived, dtype: float64

Survival Rate by Passenger Class:

Pclass

1 0.629630

2 0.472826

3 0.242363

Name: Survived, dtype: float64

===== INSIGHTS =====

1. Female passengers had a higher survival rate compared to males.
2. First-class passengers survived more than second and third class passengers.
3. Passenger fare had a positive relationship with survival.
4. Most passengers were between 20 and 40 years old.
5. Data visualization helped identify patterns and correlations effectively.

=====

KEY INSIGHTS

- Female passengers had a higher survival rate compared to male passengers.
- Passengers in first class had better survival chances.
- Age and fare influenced survival probability.
- Most passengers were between 20 and 40 years old.
- Data cleaning improved dataset quality for analysis.

CONCLUSION

The Exploratory Data Analysis performed on the Titanic dataset successfully identified important patterns and trends affecting passenger survival. The project improved understanding of data preprocessing, visualization, and statistical analysis techniques. EDA plays a significant role in preparing datasets for machine learning applications.

REFERENCES

- <https://www.kaggle.com>
- <https://pandas.pydata.org>
- <https://matplotlib.org>
- <https://seaborn.pydata.org>